

ON SOME BILHARZIA INFECTION AND IMMUNISATION MODELS

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SUMMARY

Elsdon-Dew (personal communication) suggested certain models to explain data on the incidence of Bilharzia in different age groups of a population. These models are here formulated on the basis of a point process of "potential infections", they are analysed, and the probability that an individual is manifest is calculated as function of its age.

1. Introduction

We shall use the following terminology. An individual is at any age ( $t$ ) either manifest ("it has Bilharzia") or not, it is also at any age  $t$  either immune or not. If an individual is not manifest at  $t$  but was manifest before we say it is cured at  $t$ . The term potential infections covers both effective and ineffective infections. For simplicity we shall sometimes write infection instead of potential infection. A potential infection at  $t$  is effective if the individual is not immune at  $t$ , otherwise it is ineffective. We shall write an individual has  $k$  infections at  $t$  instead of "has been exactly  $k$  times infected before or at  $t$ " and an individual becomes infected at  $t$  if an infection occurs in the interval  $(t, t + dt)$ .

Any model which intends to explain data on Bilharzia should be based on the following assumptions.

- (1) Every individual is exposed to infection, and some probability mechanism determines the time (age) points at which an individual becomes infected.
- (2) There is some rule which decides (if the rule is deterministic, or gives the probability for such a decision if the rule is probabilistic) only on the record of infections of an individual, whether the individual is manifest at a certain time  $t$ .

This basis is extremely wide and allows for an unlimited number of different models. The first part of the model is specified as soon as a stochastic point process is given which generates the infections. One could postulate a Poisson process, or another stationary point process. There is also considerable freedom in choosing the rules mentioned under (2).

We shall here analyse some of the models which were suggested by Elsdon-Dew (1966). In all of them a Poisson point process, with parameter  $\alpha$  ("endemicity"), of infections is assumed, the rules for (2) are given in the next section.

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## 2. The Poisson point process of potential infections and some sets of rules

2.1 If one assumes that an individual has at all times  $t$  the same probability  $\alpha \Delta t$  to become infected in  $(t, t + \Delta t)$ , that the occurrence of infections in  $(t, t + \Delta t)$  is independent of what happened before  $t$ , and that the probability to have more than one infection in  $(t, t + \Delta t)$  is  $o(\Delta t)$  then it follows (see e.g. Cox (1962), p. 30), that the process of infections is a Poisson process with parameter  $\alpha$ .

2.2 The following rules were suggested by Elsdon-Dew (1966). The times at which the first, second, ... infection occurs are denoted by  $\tau_1, \tau_2, \dots$

- (i) An individual becomes manifest at  $\tau_1$ . It becomes immune and cured at  $\tau_1 + c + b$ , where  $c$  is a non-negative constant, and  $b$  is an exponentially - with parameter  $\beta$  - distributed random variable, which is independent of the process of infections.
- (ii) An individual becomes manifest at  $\tau_1$  and immune at  $\tau_1 + c$ , where  $c$  is a positive constant. Between  $\tau_1$  and  $\tau_1 + c$  the individual may become effectively infected again. If the last effective infection occurs at  $\tau_k$  the individual becomes cured at  $\tau_k + d$ , where  $d \geq c$  is a constant.
- (iii) An individual becomes manifest at  $\tau_1$ . It becomes immune and cured at the time at which  $W(t)$ , its "infection time", becomes  $\omega$ , where  $\omega$  is a positive constant.

Definition of the "infection time at  $t$ ",  $W(t)$ : if an individual has  $k$  ( $k = 1, 2, \dots$ ) effective infections at  $t$  which occurred at  $\tau_1, \tau_2, \dots, \tau_k$ , then it has, at  $t$ , infection time

$$W(t) = \sum_{j=1}^k (t - \tau_j); \text{ if it has no effective infections at } t, \\ \text{then } W(t) = 0.$$

## 3. Analysis of the Models

3.1 From the assumption that the infections are generated by a Poisson process with parameter  $\alpha$  the following follows.

The inter-arrival times  $\tau_{k+1} - \tau_k$  are independently and identically distributed and have exponential distributions with parameter  $\alpha$ .

The number of infections in an interval of length  $L$  has a Poisson distribution with parameter  $\alpha L$ .

The probability density that an individual becomes for the  $k$ -th time infected at  $t$  is

$$h(t; k) = \alpha(\alpha t)^{k-1} e^{-\alpha t} / (k-1)! .$$

The probability to have at least  $k$  infections ( $k = 1, 2, \dots$ ) at  $t$  is

$$H(t; k) = \int_0^t h(\tau; k) d\tau = (\alpha t)^k e^{-\alpha t} {}_1F_1(1; k+1; \alpha t) / k!,$$

where  ${}_1F_1$  is the confluent hypergeometric function.

The probability to have no infection at  $t$  is

$$1 - H(t; 1) = e^{-\alpha t}.$$

The probability to have exactly  $k$  infections at  $t$  is

$$H(t; k) - H(t; k + 1) = (\alpha t)^k e^{-\alpha t} / k! .$$

For every  $t$  the probability density to have become the last time infected at  $\tau$ ,  $\tau \leq t$ , is

$$\alpha(1 - H(t - \tau; 1)) = \alpha e^{-\alpha(t-\tau)}.$$

At  $t$ , the probability density to become for the first time infected at  $\tau$  and for the last time at  $\tau + \sigma$  is

$$\begin{aligned} g(\tau, \sigma; t) &\equiv g(\sigma; t) \\ &= \alpha h(\tau; 1)(1 - H(t - \sigma - \tau; 1)) = \alpha^2 e^{-\alpha(t-\sigma)} \\ &\quad \text{for } 0 < \sigma \leq t - \tau \\ &= h(\tau; 1)(1 - H(t - \tau; 1)) = \alpha e^{-\alpha t} \quad \text{for } \sigma = 0. \end{aligned}$$

The probability density to become for the first time infected at  $\tau$ , to become also infected at  $\tau + \sigma$ , and not to be infected between  $\tau + \sigma$  and  $\tau + \rho$ , ( $\sigma \leq \rho$ ), is  $g(\sigma; \tau + \rho)$ .

3.2 Call  $I(t)$  the probability to be immune at  $t$ ,  $C(t)$  the probability to be cured at  $t$ , and  $M(t)$  the probability to be manifest at  $t$ .

### 3.2.1 Model 2.2(i)

The probability of an individual to be immune at  $t$ , given that it became for the first time infected at  $\tau$ ,  $\tau < t$ , is

$$1 - e^{-\beta(t-\tau-c)} \text{ for } c \leq t - \tau,$$

and zero for  $t - \tau \leq c$ .

Thus

$$\begin{aligned} I(t) &= 0 && \text{for } t \leq c \\ &= \int_0^{t-c} h(\tau; 1)(1 - e^{-\beta(t-\tau-c)}) d\tau && \text{for } c \leq t. \end{aligned}$$

And because  $C(t) = I(t)$ , the probability to be manifest at  $t$  is

$$\begin{aligned} M(t) &= H(t; 1) - C(t) \\ &= e^{-\alpha t}(e^{\alpha c} - 1) + \alpha(e^{-\beta(t-c)} - e^{-\alpha(t-c)})/(\alpha - \beta) \\ &\quad \text{for } c \leq t \text{ and } \alpha \neq \beta \\ &= e^{-\alpha t}(e^{\alpha c} - 1) + \alpha(t-c)e^{-\alpha(t-c)} && \text{for } c \leq t \text{ and } \alpha = \beta \\ &= 1 - e^{-\alpha t} && \text{for } t \leq c. \end{aligned}$$

### 3.2.2 Model 2.2(ii)

The probability density to become for the first time infected at  $\tau$  and for the last time effectively infected at  $\tau + \sigma$  is  $g(\sigma; \tau + c)$ .

The probability density to be the last time effectively infected at  $t$ ,  $\ell(t; c)$ , is then, for  $c \leq t$

$$\begin{aligned} \ell(t; c) &= \int_{t-c}^t g(t-\tau; \tau+c) d\tau + g(0; t+c) \\ &= (\alpha/2) e^{-\alpha(t+c)} (1 + e^{2\alpha c}). \end{aligned}$$

For  $t \leq c$  one has

$$\begin{aligned} \ell(t; c) &= \int_0^{t-c} g(t-\tau; \tau+c) d\tau + g(0; t+c) \\ &= (\alpha/2) e^{-\alpha(t+c)} (1 + e^{2\alpha t}) \end{aligned}$$

The probability density to become cured at  $t$ ,  $k(t; c, d)$ , say, is then

$$\begin{aligned} k(t; c, d) &= 0 && \text{for } 0 \leq t \leq d \\ &= \ell(t-d; c) && \text{for } d \leq t. \end{aligned}$$

Therefore

$$\begin{aligned} C(t) &= \int_0^t k(\tau; c, d) d\tau \\ &= 0 && \text{for } 0 \leq t \leq d \\ &= (1/2) e^{-\alpha c} (e^{\alpha(t-d)} - e^{-\alpha(t-d)}) && \text{for } d \leq t \leq c + d \\ &= 1 - (1/2) e^{-\alpha(t-d)} (e^{\alpha c} + e^{-\alpha c}) && \text{for } c + d \leq t, \end{aligned}$$

and

$$\begin{aligned} M(t) &= H(t; 1) - C(t) \\ &= 1 - e^{-\alpha t} && \text{for } 0 \leq t \leq d \\ &= 1 - e^{-\alpha t} - (1/2) e^{-\alpha c} (e^{\alpha(t-d)} - e^{-\alpha(t-d)}) && \text{for } d \leq t \leq c + d \\ &= e^{-\alpha t} ((e^{\alpha(c+d)} + e^{-\alpha(c-d)})/2 - 1) && \text{for } c + d \leq t. \end{aligned}$$

### 3.2.3 Model 2.2(iii)

Here one obviously has

$$M(t) = P(0 < W(t) \leq \omega).$$

The problem is therefore to find the distribution function of  $W(t)$ . We give two methods to solve it, one is using the fact that  $W(t)$  is the integrated Poisson counting process, the other is based on a conditional argument.

3.2.3.1 If one denotes the Poisson counting process with parameter  $\alpha$ , by  $N(t)$ ,  $t \geq 0$ , then

$$W(t) = \int_0^t N(u) du.$$

There is a well-known equation relating the moments of an integrated process to the moments of the original process

$$\begin{aligned} \mu_h^i(W(t)) &= \int_0^t \dots \int_0^t \mu_{1, \dots, 1}^i(N(u_1), N(u_2), \dots, N(u_h)) \\ &\quad du_1 du_2 \dots du_h. \end{aligned}$$

In the case of a Poisson process this equation is not very useful because the expressions for  $\mu_{1, \dots, 1}^i$  are for large  $h$  very cumbersome. The cumulants  $\kappa_{1, \dots, 1}^i$  are, however, simple and there is another equation, which is analogous to the one above, and which can easily be proved:-

$$\begin{aligned} \kappa_h(W(t)) &= \\ &= \int_0^t \dots \int_0^t \kappa_{1, \dots, 1}^h(N(u_1), N(u_2), \dots, N(u_h)) du_1 du_2 \dots du_h \\ &= h! \int_0^t \dots \int_0^t \kappa_{1, \dots, 1}^h(N(u_1), N(u_2), \dots, N(u_h)) du_1 du_2 \dots du_h, \end{aligned}$$

where the integral is over  $0 \leq u_1 \leq u_2 \leq \dots \leq u_h \leq t$ .

The cumulants  $\kappa_{1, \dots, 1}^i$  can be obtained from the joint distribution of  $N(u_i) = x_i$ , say,  $(i=1, 2, \dots, h)$ .

Assume  $u_i > u_{i-1}$ , write  $u_i = \sum_{j=1}^i v_j$ ,  $v_j > 0$  and  $y_i = x_i - x_{i-1}$ ,

$(i=1, 2, \dots, h)$ , where  $x_0 = 0$ . Then the  $y_i$  are independently distributed, each having a Poisson distribution with parameter  $\alpha v_i$ , and

$$E(\exp(i \sum_{j=1}^h y_j s_j)) = \exp(\alpha \sum_{j=1}^h v_j (e^{is_j} - 1)).$$

The ch.f. of the joint distribution of  $x_i (i=1, 2, \dots, h)$  is

$$E(\exp(i \sum_{j=1}^h x_j t_j)) = E(\exp(i \sum_{j=1}^h \sum_{k=1}^j y_k t_j))$$

$$\begin{aligned}
&= E(\exp(i \sum_{j=1}^h y_j \sum_{k=j}^h t_k)) \\
&= \exp(\alpha \sum_{j=1}^h v_j (\exp(i \sum_{k=j}^h t_k) - 1)).
\end{aligned}$$

The cumulants  $\kappa_{11\dots 1}$  are obtained by differentiating the logarithm of the ch.f.:

$$\kappa_{11\dots 1} = \alpha v_1 = \alpha u_1.$$

Above it was assumed that  $u_i > u_{i-1}$ .

If this is not necessarily so, one has

$$\kappa_{11\dots 1}(N(u_1), N(u_2), \dots, N(u_h)) = \alpha \min_i u_i.$$

One may prove by induction that

$$\int_0^t \int_0^{u_h} \dots \int_0^{u_2} u_1 \, du_1 du_2 \dots du_h = \frac{t^{h+1}}{(h+1)!}.$$

Therefore

$$\kappa_h(W(t)) = t^{h+1}/(h+1),$$

and the cumulant generating function of  $W(t)$  is

$$\psi_{W(t)}(s) = (\alpha/is)(e^{ist} - 1 - ist).$$

### 3.2.3.2

We consider now directly  $W(t) = \sum_{j=1}^k (t - \tau_j)$ , where  $k$  has a Poisson distribution with parameter  $\alpha t$ . Conditionally, given that

$k(k=1,2,\dots)$  events have occurred before or at  $t$ , the  $\tau_i$  are - if the order in the indices is neglected - uniformly and independently distributed over the interval  $(0,t)$ . If the  $\tau_i$  have this distribution then the  $t-\tau_i$  have the same distribution.

It follows straight away that  $W(t)$  has a compound Poisson distribution, associated with the uniform  $(0,t)$  distribution. Thus the cumulant generating function of  $W(t)$  is

$$\begin{aligned}
\psi_{W(t)}(s) &= \alpha t (\phi_{t-\tau_i}(s) - 1) \\
&= (\alpha/is) (e^{ist} - 1 - ist),
\end{aligned}$$

which agrees with the result obtained in 3.2.3.1.

The same argument gives also the p.d.f. - or the d.f. if one wishes - of  $W(t)$ . The distribution of a sum of independently and uniformly distributed random variables has apparently at first been obtained by Laplace. (See e.g. Feller (1946)). It is now sometimes called the Irwin-Hall distribution after some of its rediscoverers (Irwin, (1927); Hall, (1927)).

One has then

$$\begin{aligned}
&P(0 \leq W(t) \leq \omega \mid k) \\
&= \sum_{j=0}^s (-1)^j t^{-k} (\omega - jt)^k / j! (k-j)! && \text{for } k=1,2,\dots \\
& && \text{and } st \leq \omega \leq (s+1)t, \\
& && s=0,1,\dots,k-1 \\
&= 1 && \text{for } k=0,1,\dots \\
& && \text{and } kt \leq \omega
\end{aligned}$$

Thus

$$P(0 \leq W(t) \leq \omega) = \sum_{k=0}^{\infty} e^{-\alpha t} \frac{(\alpha t)^k}{k!} P(0 \leq W(t) \leq \omega \mid k)$$

$$= e^{-\alpha t} \left( \sum_{k=0}^s \frac{(\alpha t)^k}{k!} + \sum_{k=s+1}^{\infty} \frac{\alpha^k}{k!} \sum_{j=0}^s (-1)^j (\omega - jt)^{k/j!(k-j)!} \right) \quad \text{for } st \leq \omega \leq (s+1)t$$

$s=0, 1, \dots,$

and  $M(t) = P(0 < W(t) \leq \omega)$

$$= e^{-\alpha t} \left( \sum_{k=0}^s \frac{(\alpha t)^k}{k!} - 1 + \sum_{k=s+1}^{\infty} \frac{\alpha^k}{k!} \sum_{j=0}^s (-1)^j (\omega - jt)^{k/j!(k-j)!} \right) \quad \text{for } st < \omega \leq (s+1)t$$

$s=0, 1, \dots,$

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